

TWEET ANALYSIS DASHBOARD

Data Limitations & Adjustments Report

Power BI Implementation | Internship Project | 2026

Overview

This document outlines the data limitations identified during the implementation of the Tweet Analysis Dashboard in Power BI. Each task was analyzed against the actual dataset (1,166 clean rows, June–October 2020) and adjustments were made where the original task requirements could not be met due to data constraints. All decisions are documented here for transparency.

TASK 1

Tweet Interaction Breakdown by Category

Status: IMPLEMENTED WITH ADJUSTMENT

Word Count Filter Adjusted

The task required filtering tweets with word count > 40. However, no tweet in this dataset exceeds 36 words. Applying this filter returns 0 results. The threshold was adjusted to > 20 to preserve analytical intent while ensuring the visualization shows meaningful data.

Time Visibility Applied

Chart is visible only between 3 PM and 5 PM IST using the IsVisible_3PM_5PM DAX measure set as a visual-level filter.

Even Date Filter Applied

TweetDay column was created in Power Query and even-numbered days (2,4,6...30) are filtered using basic filtering.

TASK 2

Engagement Rate: App Opens vs No App Opens

Status: PARTIALLY IMPLEMENTED — DATA LIMITATIONS

App Opens Data Extremely Sparse

Only 2 tweets out of 1,166 have app opens > 0. App installs column is entirely zero across all rows. The comparison is heavily imbalanced but the chart was built to reflect actual data distribution. This imbalance is itself a finding.

TweetHourIST Filter Removed

The task required filtering tweets posted between 9 AM and 5 PM. However, the 2 tweets with app opens were posted outside this window. Applying the filter removes both, leaving the "With App Opens" bar empty. Filter was removed to preserve the visual.

HasNoLetterD Filter Removed

CONTAINSSTRING in DAX is case-insensitive, so filtering for the letter "D" also removes tweets containing lowercase "d". This eliminates almost all data. The filter was removed to maintain chart integrity. Documented here as a known limitation.

IsEvenImpressions Filter Removed

The 2 tweets with app opens happen to have odd impressions. Applying the even impressions filter removes both, leaving the "With App Opens" bar completely empty. Filter was removed to preserve the comparison.

**TAS
K 3**

Media Interaction by Day of Week

Status: IMPLEMENTED WITH NOTES

Dual-Axis Chart Correctly Built

Unlike the senior dashboard which used Tweet ID on the X-axis, this implementation correctly uses Day of Week (Monday through Sunday). DayOfWeekNum column was created to sort days in correct order.

HasNoLetterH Filter Not Applied

Same issue as letter "D" in Task 2. CONTAINSSTRING is case-insensitive and removes tweets containing lowercase "h", which eliminates the majority of the dataset. Filter was omitted and documented here.

CharCount Filter Behaviour

All 1,166 tweets have a character count greater than 30, so this filter passes all data. It was added for compliance but has no practical effect on the visualization.

Time Visibility Applied

Chart is visible between 3 PM to 5 PM IST and 7 AM to 11 AM IST using IsVisible_Task3 DAX measure.

**TAS
K 4**

Replies, Retweets and Likes Comparison

Status: FULLY IMPLEMENTED — NO LIMITATIONS

All Requirements Met

Bar chart showing Sum of Replies (666), Sum of Retweets (375), and Sum of Likes (5,747) for the June to August 2020 period. 753 tweets fall within this date range. No complex filters were required for this task.

**TAS
K 5**

Monthly Engagement Rate Trend

Status: IMPLEMENTED WITH ADJUSTMENT

Dataset Does Not Cover Full Year

The task asks for a trend across all 12 months of the year. However, the dataset only contains data from June to October 2020 (5 months). The chart was built with the available data and retitled to "Monthly Engagement Rate Trend (Jun-Oct 2020)" to accurately reflect the actual data range.

Two Lines Correctly Implemented

MediaFlag calculated column was created to split tweets into "Media" and "No Media" categories based on media views > 0. The line chart correctly shows two separate trend lines with the legend.

TASK 6

Top 10 Tweets by Engagement

Status: IMPLEMENTED WITH ADJUSTMENT

No Username or Profile Column in Dataset

The task requires displaying the "user profile associated with each tweet." However, the dataset contains no username or profile name column. Only a numeric Tweet ID column exists. Tweet ID was used as the identifier with a note added to the dashboard.

Word Count Filter Applied

Word count < 30 filter was applied. 768 tweets qualify after this filter, giving a valid pool of 192 tweets after all other filters are applied. Top 10 are selected from this pool.

All Other Filters Applied

IsWeekday = True (excludes weekends), TweetDay odd numbers, IsEvenImpressions = 1, and IsVisible_3PM_5PM = 1 (visible only 3 PM to 5 PM IST) are all applied correctly.

Summary of All Adjustments

Task	Issue	Action Taken
Task 1	Word count > 40 = 0 tweets	Adjusted threshold to > 20
Task 2	App opens data: only 2 tweets	Chart built, imbalance noted
Task 2	TweetHourIST removes app opens data	Filter removed, documented
Task 2	Letter D filter removes all data	Filter removed, documented
Task 2	Even impressions removes app opens	Filter removed, documented
Task 3	Letter H filter removes majority	Filter omitted, documented

Task 3	Colleague used wrong axis (Tweet ID)	Fixed: Day of Week used correctly
Task 4	No issues	Fully implemented
Task 5	Only 5 months of data not 12	Chart retitled Jun-Oct 2020
Task 6	No username column in dataset	Tweet ID used as identifier

All limitations were identified through thorough data analysis before building the dashboard. Adjustments were made to ensure maximum analytical value while maintaining data integrity. Each decision is documented to demonstrate a professional understanding of the dataset.